Lecture 1 (Week 1)
Chapter 1

Exploring the Multidisciplinary Nature of Interaction Design

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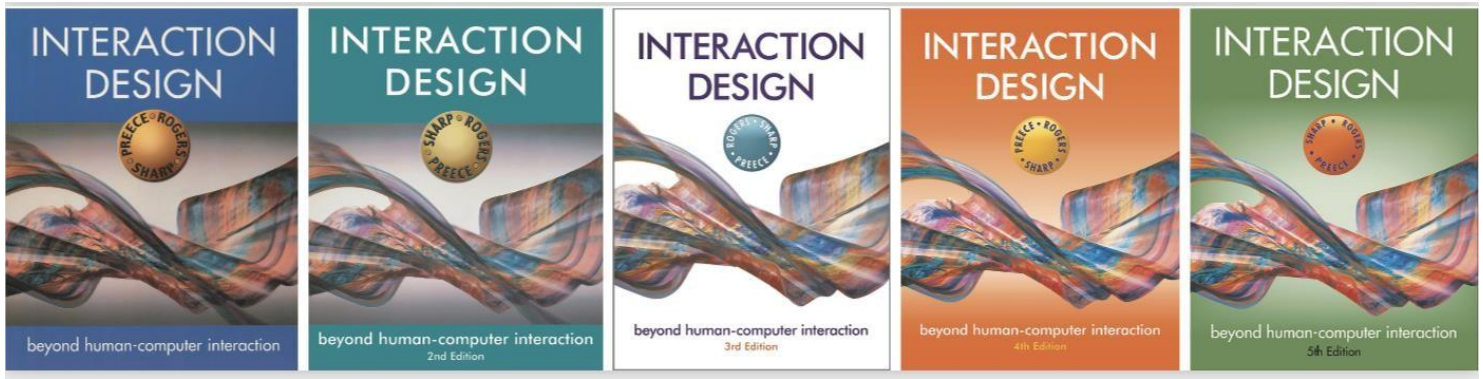
Textbook

Yvonne Rogers, Helen Sharp, and Jenny Preece (6th Edition)



Bad Designs

- Elevator controls and labels on the bottom row all look the same, so it is easy to push a label by mistake instead of a control button.
- People do not make same mistake for the labels and buttons on the top row. Why not?

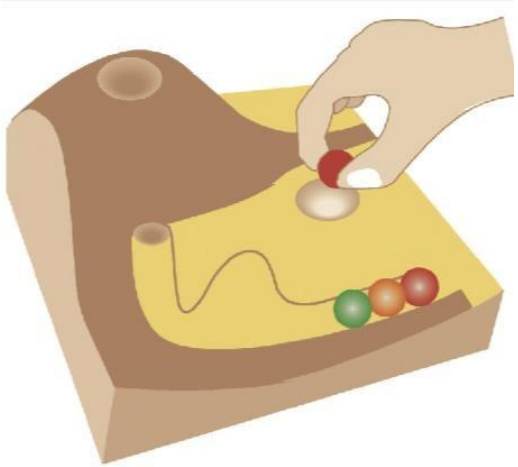
www.baddesigns.com

Why is this vending machine so bad?

- Need to push button first to activate reader
- Normally insert bill first before making selection

Good Design

- Marble answering machine (Bishop,1995)
- Based on how everyday objects behave
- Easy, intuitive, and a pleasure to use
- Only requires one-step actions to perform core tasks



Good and Bad Design

Why is the TiVo remote much better designed than standard remote controls?

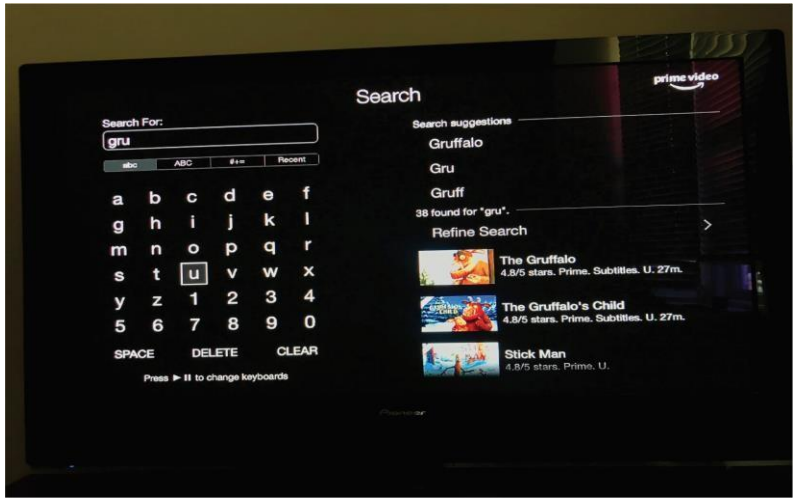
- Peanut shaped to fit in hand
- Logical layout and color-coded, distinctive buttons
- Easy-to-locate buttons

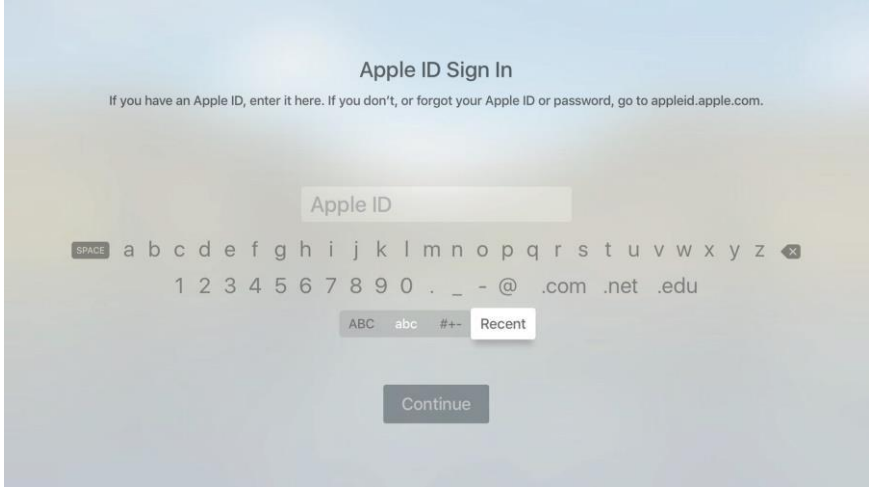
The question is how did they come up with such a good design?

Dilemma

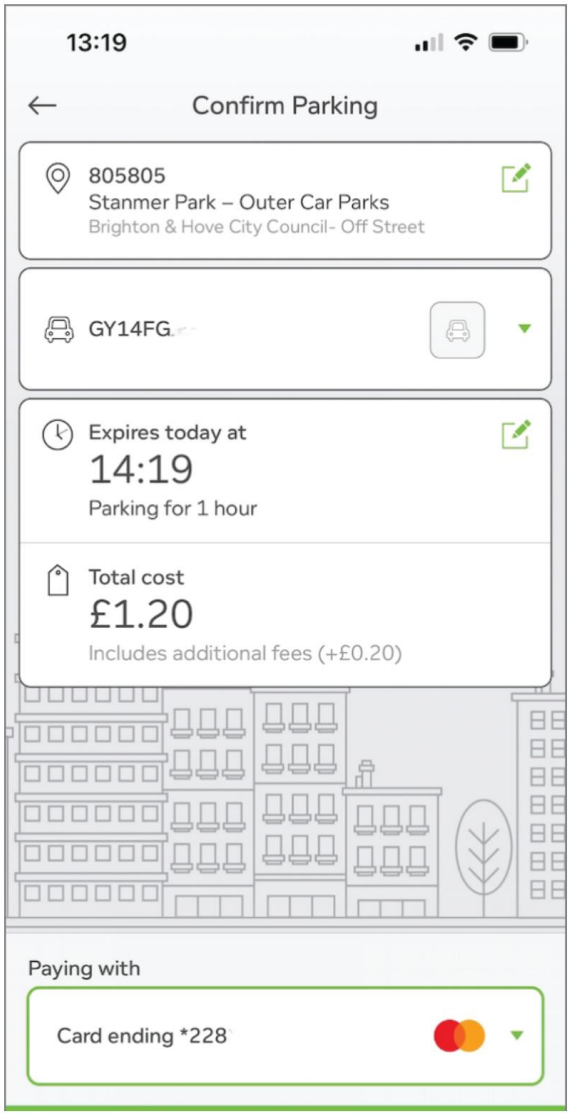
Which is the best way to interact with a smart TV? Why?

- Pecking using a grid keyboard via a remote control
- Swiping across two alphanumeric rows using a touchpad on a remote control
- Voice control using remote or smart speaker





Switching to Digital



What to Design

- Need to take into account:
  - Who the users are
  - What activities are being carried out
  - Where interaction is taking place

- Need to optimize user interactions
  - So that they match the users activities and needs

What is Interaction Design?

- "**Designing interactive products** to support the way **people communicate and interact** in their **everyday and working lives.**"

  Sharp,Rogers, and Preece (2019)

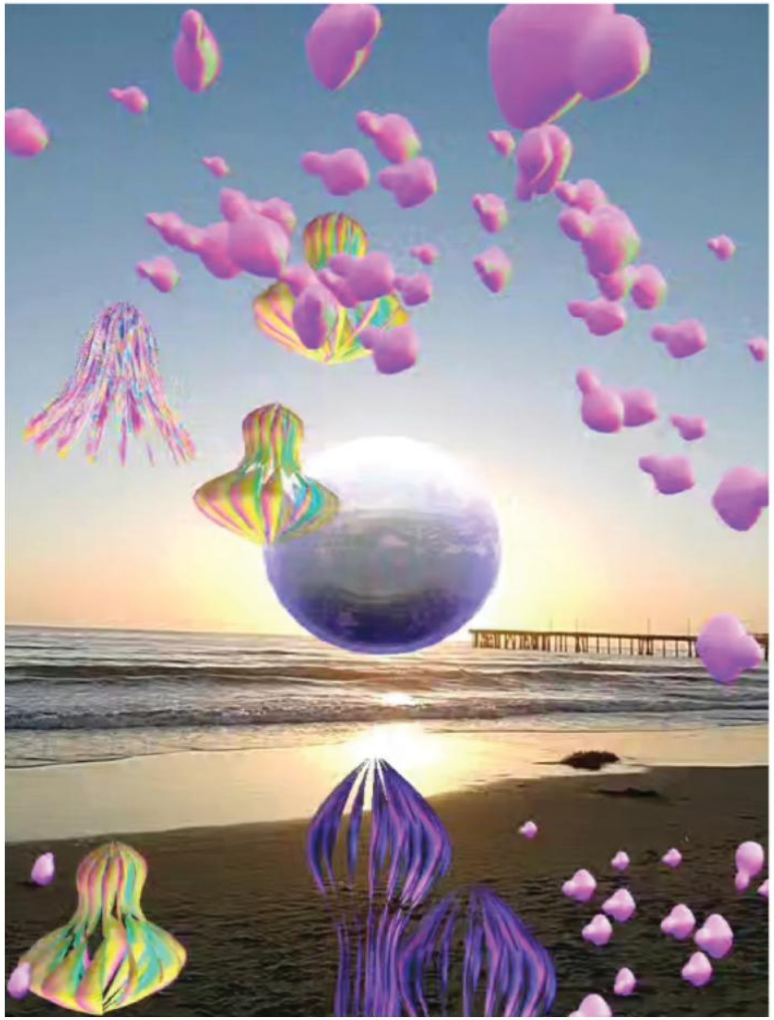
- "The **design** of **spaces** for **human communication and interaction.**"

  Winograd (1997)

Goals of Interaction Design

- Develop **usable** products
  - Usability means easy to learn, effective to use, and provides an enjoyable experience
- Involve users in the design process

User Experience Example



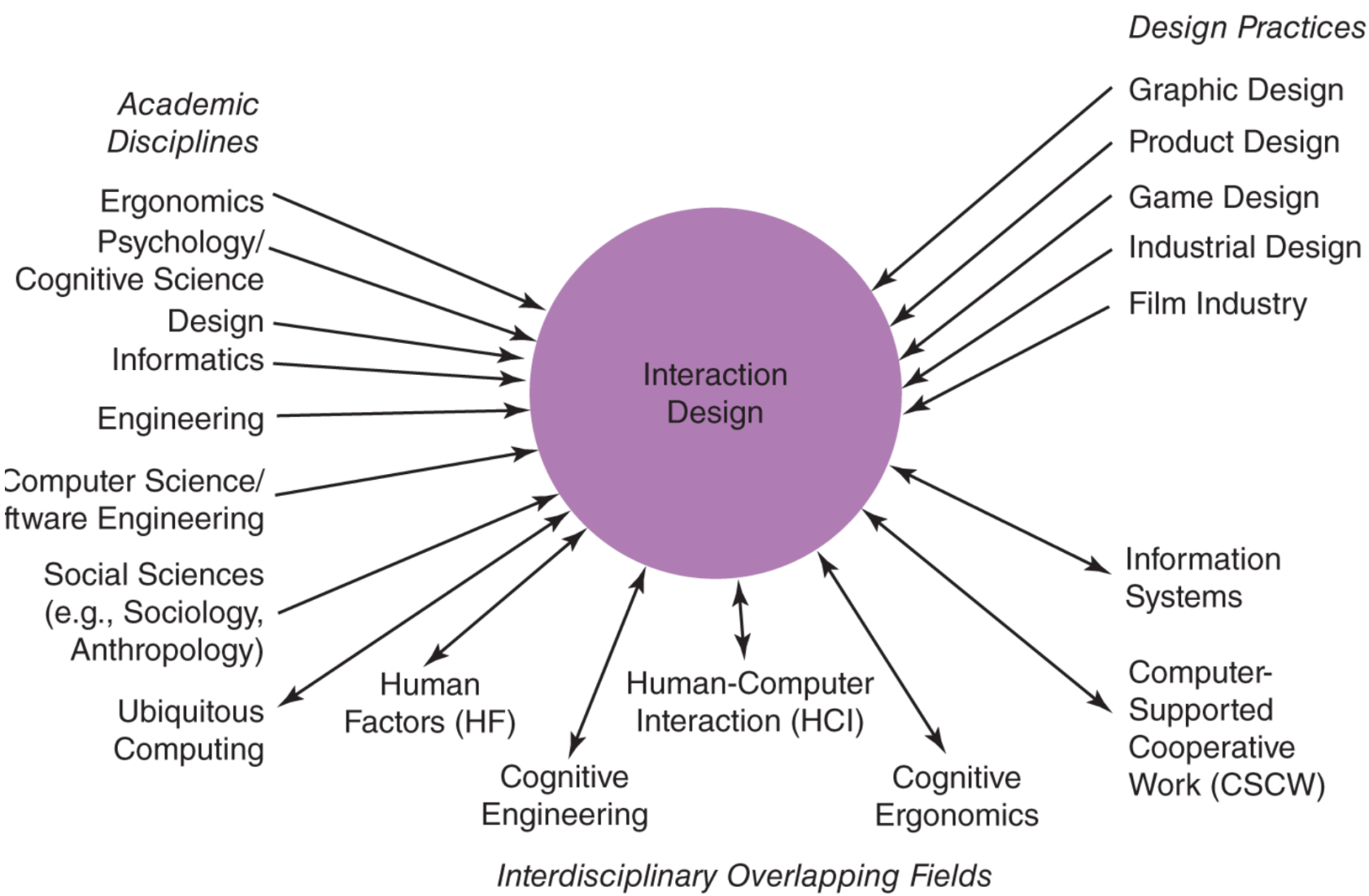
The digital world overlaying the physical experienced when wearing Snap AR Spectacles

What Kind of Design?

- Number of other terms used to emphasize what is being designed, for example:
  - User interface design, software design, user-centered design, product design, web design, experience design (UX)

- Interaction design is the umbrella term covering all of these aspects:
  - Fundamental to all disciplines, fields, and approaches concerned with researching and designing computer-based systems for people

Interaction Design (ID)



Academic Disciplines

- Academic disciplines contributing to ID:
  - Psychology
  - Social Sciences
  - Computing Sciences
  - Engineering
  - Ergonomics
  - Informatics

Design Practices

- Design practices contributing to ID:
  - Graphic design
  - Product design
  - Artist-design
  - Industrial design
  - Film industry

Interdisciplinary Fields

- Interdisciplinary fields that 'do' interaction design:
  - HCI
  - Ubiquitous Computing
  - Human Factors
  - Cognitive Ergonomics
  - Information Systems

Since we first created Figure on slide 14, many other computer-related fields have emerged where the user is considered central. Would it make sense to add these, and, if so, how? Can you also name some new fields?

Working in Multidisciplinary Teams

- Many people from different backgrounds involved
- Different perspectives and ways of seeing and talking about things
- **Benefits:** More ideas and designs generated
- **Disadvantages:** Difficult to communicate and progress forward the designs being create

Interaction Design in Business

Nielsen Norman Group:"help companies enter the age of the consumer, designing human-centered products and services"

Cooper: "From research and product to goal-related design"

IDEO:"creates products, services and environments for companies pioneering new ways to provide value to their customers"

The User Experience

- How a **product behaves** and is **used by people** in the real world
  - The way people **feel** about it and their pleasure and satisfaction when using it, looking at it, holding it, and opening or closing it
  - "Every product that is used by someone has a user experience: newspapers, ketchup bottles, reclining armchairs, cardigan sweaters." (Garrett, 2010)
  - "All aspects of the end-user's interaction with the company, its services, and its products. (Nielsen and Norman, 2014)

- Cannot design a user experience—only can design for a user experience

Defining User Experience

- How users **perceive** a product:

such as whether a smartwatch is seen as sleek or chunky, and their emotional reaction to it, such as whether people have a positive experience when using it (Hornbæk and Hertzum, 2017)

- Hassenzahl's (2010) model of the user experience
  - Pragmatic:
    - how **simple, practical, and obvious** it is for the user to achieve their goals
  - Hedonic:
    - how **stimulating** the interaction is to users

iPod User Experience: A Success?

- Quality user experience from the start

- Simple, elegant, distinct brand, pleasurable, must have fashion item, catchy names, cool

Figure 1.6 The iPod Nano Touch
Source: ©Press Association, reproduced with permission.

Characteristics of Interaction Design

- **Users** should be involved throughout the development of the project
- Specific usability and user experience **goals** need to be identified, clearly **documented**, and agreed to at the beginning of the project
- **Iteration** is needed through the core activities

Usability Goals

- Effective to use
- Efficient to use
- Safe to use
- Have good utility
- Easy to learn
- Easy to remember how to use

User Experience Goals

Desirable aspects		
Satisfying	Helpful	Fun
Enjoyable	Motivating	Provocative
Engaging	Challenging	Surprising
Pleasurable	Enhancing sociability	Rewarding
Exciting	Supporting creativity	Emotionally fulfilling
Entertaining	Cognitively stimulating	Experiencing flow
Undesirable aspects		
Boring	Unpleasant	Creepy
Frustrating	Patronizing	Intrusive
Making one feel guilty	Making one feel stupid	Invasive
Annoying	Cutesy	Deceptive
Childish	Gimmicky	Annoying